

IVDEL', Russian Federation

WMO: 239210

Lat: 60.683N

Lon: 60.450E

Elev: 309

StdP: 14.53

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-28.8	-38.7	0.6	-33.6	-33.8	0.8	-28.7	14.0	18.1	11.8	15.7	1.5	320	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	84.2	67.8	80.5	66.1	76.8	63.6	69.8	80.3	67.7	78.0	65.5	74.4	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.5	74.4	63.8	89.7	72.7	61.8	83.5	70.1	34.0	80.5	32.3	78.2	30.5	74.5	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.6	9.9	-39.0	88.8	5.8	2.9	-43.2	90.9	-46.6	92.6	-49.9	94.3	-54.1	96.4		
(5)			-39.1	73.1	5.7	2.5	-43.3	74.9	-46.6	76.4	-49.8	77.8	-54.0	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.1	-1.5	3.7	20.9	34.9	47.0	58.1	63.1	57.0	46.5	34.4	15.2	3.8		
	DBStd	24.78	14.89	14.68	10.88	9.14	8.65	8.11	7.13	7.07	7.58	8.66	15.60	16.67		
	HDD50	7571	1596	1295	902	457	165	21	3	14	155	487	1045	1431		
	HDD65	12122	2061	1715	1367	904	558	234	124	265	555	948	1495	1896		
	CDD50	1034	0	0	0	3	74	265	407	231	51	3	0	0		
	CDD65	111	0	0	0	0	1	28	64	17	1	0	0	0		
	CDH74	1400	0	0	0	0	75	361	743	202	19	0	0	0		
	CDH80	345	0	0	0	0	9	81	200	53	2	0	0	0		
	WSAvg	3.8	2.9	3.3	4.6	4.9	4.8	4.1	3.3	3.3	3.6	4.2	3.5	3.1		
	PrecAvg	21.5	1.1	0.7	1.1	1.3	2.1	2.8	2.8	3.0	2.8	1.5	1.3	0.9		
(15) Precipitation	PrecMax	28.5	2.0	1.5	2.8	2.2	4.2	6.4	6.0	6.6	6.3	2.9	2.2	1.7		
	PrecMin	13.7	0.2	0.1	0.0	0.0	0.4	0.8	0.8	1.3	0.3	0.4	0.1	0.2		
	PrecStd	3.1	0.5	0.4	0.7	0.6	1.0	1.6	1.2	1.6	1.6	0.8	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	38.3	48.4	67.7	81.1	86.2	88.7	86.3	76.6	61.1	43.9	38.9		
		MCWB	31.0	33.1	38.3	52.3	60.7	66.9	69.9	70.3	62.7	51.5	39.1	34.2		
	2%	DB	28.5	33.6	43.1	59.9	75.4	82.2	85.3	79.1	67.9	55.3	40.1	34.8		
		MCWB	26.0	29.9	35.6	47.2	56.9	65.0	68.8	65.8	57.4	48.9	36.3	31.4		
	5%	DB	24.8	29.6	39.5	54.7	70.3	78.1	81.9	74.3	62.9	50.5	37.1	30.9		
		MCWB	22.7	26.4	33.2	43.8	54.7	63.4	67.5	63.7	55.2	45.4	33.5	28.3		
	10%	DB	20.6	25.6	36.4	50.0	65.1	73.8	78.3	70.1	59.0	46.5	34.2	26.8		
		MCWB	18.7	23.0	31.3	41.0	52.0	60.9	66.0	61.4	52.9	42.6	31.2	24.5		
	0.4%	WB	31.3	33.4	39.6	53.0	61.7	70.3	73.2	71.3	63.6	52.9	39.9	34.4		
		MCDB	33.6	37.4	46.5	66.1	76.6	80.5	82.8	81.2	73.9	58.6	42.9	37.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.2	30.3	36.2	48.0	58.6	67.2	70.7	67.5	58.7	49.0	36.8	31.4		
		MCDB	28.1	33.7	41.6	58.4	72.0	78.3	81.1	77.4	65.9	54.1	39.7	34.6		
	5%	WB	22.8	26.8	33.7	44.5	56.0	65.0	69.0	64.8	56.0	45.8	33.7	28.3		
		MCDB	24.8	29.6	38.8	53.8	67.7	75.4	79.3	72.1	61.8	50.2	36.5	30.9		
	10%	WB	18.7	23.0	31.6	41.6	53.2	62.8	67.2	62.3	53.6	42.9	31.3	24.6		
		MCDB	20.6	25.7	36.3	48.6	63.5	71.4	76.9	68.9	58.4	46.2	33.9	26.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	19.5	19.9	18.4	21.4	19.6	19.8	17.6	15.4	11.8	12.4	14.8		
		MCDBR	16.0	18.0	20.8	28.1	32.2	26.8	26.7	25.3	23.0	18.5	11.4	13.4		
		MCWBR	14.3	15.0	14.6	16.8	16.1	12.3	12.6	13.3	13.4	12.5	9.2	11.5		
		MCDBR	15.8	16.9	18.6	26.1	28.5	23.6	23.9	23.4	20.5	17.4	10.7	13.3		
	5% WB	MCWBR	14.3	14.4	13.6	16.3	15.5	12.1	12.4	13.2	12.2	12.2	9.1	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.294	0.343	0.379	0.380	0.385	0.413	0.407	0.353	0.319	0.268	0.237		
	taud		2.021	2.069	2.033	2.133	2.340	2.384	2.304	2.297	2.422	2.363	2.119	1.954		
	Ebn at Noon		148	212	239	255	267	268	255	246	242	209	143	99		
	Edh at Noon		14	25	36	40	35	34	36	34	25	19	13	9		
(44) All-Sky Solar Radiation	RadAvg		114	363	743	1164	1551	1632	1656	1200	689	344	130	60		
	RadStd		9	40	64	146	137	134	138	114	102	41	17	6		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page